Foldable but not dockable: when message passing over-squashes at protein–protein interfaces, and when it does not (a PINDER study)

Graph Representation Learning mini-project report; **Murat Kolic**, matrikelnummer: 12551216.
Code: `https://github.com/muuki2/grl-oversquashing-pinder`

Abstract. AlphaFold folds individual chains confidently yet often mis-places the *complex*. Taking this "great monomers, weak complexes" regime as *motivation*, we ask a message-passing question: *when* does *over-squashing* govern protein–protein interface inference? We treat the interface as a sparse *inter*-chain bottleneck a network must route through, studying it with a GNN we *control*, on PINDER's leakage-free split, with AlphaFold-predicted monomers (+pLDDT) as realistic *pose-free* input and the bound complex only for labels (*no folding inference*). We do not claim AlphaFold itself over-squashes, only that sparse message passing over the interface does. A synthetic probe establishes the mechanism cleanly (ρ=−0.97/+0.97 for resistance/Jacobian vs. accuracy). Our central real-data result is that over-squashing at the interface is **contingent on node-feature richness**: varying *only* the per-residue features from weak (degree+pLDDT, one-hot) to rich (ESM2), the benefit of cross-chain rewiring ΔAUPRC(knn−none) *flips sign* from +0.086 to −0.017; it relieves a genuine bottleneck when features are weak, but mildly *over-smooths* once rich features let the model solve the interface locally. In the weak regime the mechanism is causal: every rewiring beats the disconnected baseline (+0.03–0.04 AUPRC, ECE 0.084→0.038; a curvature-targeted SDRF matches them adding only ∼48 edges/system), and the trained-model Jacobian rises from 0 to $\sim 5\times 10^{-2}$. Per-system failure tracks the *modelled* resistance ($\rho = -0.49$, confirmed within-system $\rho = -0.13$) but not the pose-unseen *native* one ($\rho \approx -0.11$): a principled dissociation for *pose-free* inputs. We report bootstrap CIs, one-sided tests, and a minimum-detectable-effect throughout.

1 Introduction

AlphaFold(-Multimer) produces confidently folded monomers but a large fraction of its *complex* predictions are wrong, the documented "great monomers, weak complexes" regime (Bryant et al., 2022; Evans et al., 2022). Why should placing two well-folded chains relative to one another be so much harder than folding each? We propose a graph-theoretic reframing grounded in a core property of message-passing GNNs: *over-squashing*, the exponential compression of information forced through a graph bottleneck (Alon & Yahav, 2021; Topping et al., 2022; Di Giovanni et al., 2023). Modelled as a graph, a folded chain is internally dense, so intra-chain reasoning has *no* bottleneck; the interface between two chains is a thin neck of a few contacts through which all cross-chain signal must squeeze, precisely the topology in which message passing over-squashes.

Research question. *Is the difficulty of inferring a protein–protein interface governed by the graph bottleneck between the two chains (an over-squashing effect) rather than by how confidently each chain is folded, and does this survive on PINDER's leakage-free split?* This tests whether a *course concept*, information propagation / over-squashing, *quantitatively* explains a real failure mode, and whether the textbook *fix* (rewiring) transfers; we *measure* the relevant properties (effective resistance, Jacobian) rather than merely invoking them.

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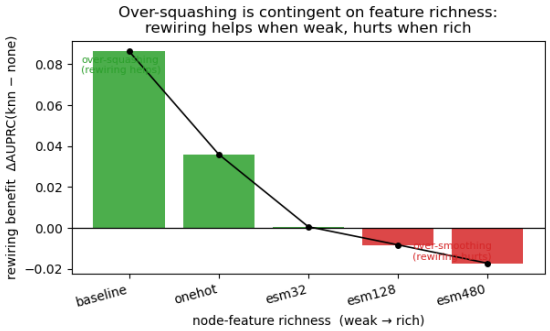
2 Methods and experimental setup

Design and diagnostics. We use a two-tier *synthetic probe → real validation* structure with the *same* two over-squashing diagnostics threading both tiers, so the real-data metric is calibrated against ground truth: the *effective resistance* $R_{\text{eff}}(u,v)=(\boldsymbol{e}_u-\boldsymbol{e}_v)^\top L^+(\boldsymbol{e}_u-\boldsymbol{e}_v)$ across the interface, the canonical bottleneck measure for over-squashing (Black et al., 2023); and the *Jacobian sensitivity* $\|\partial \boldsymbol{h}_v^{(T)}/\partial \boldsymbol{x}_u\|$ of a read-out node v to a source input u (Topping et al., 2022), measured both over freshly initialised models (an architecture property) and *across the interface on the trained real models*. The *synthetic probe* joins two Erdős–Rényi communities (16 nodes each) by exactly k cross-edges and reads a B-source label out at the A nodes, so the task *requires* cross-interface flow; we sweep $k \in \{1,2,4,8,16\}$ and depth $\in \{2,4\}$ (Supplementary Fig. 3).

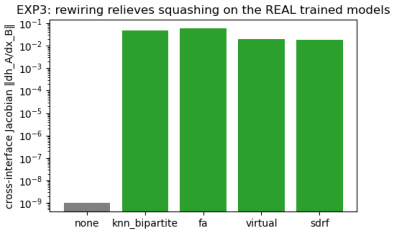
Real study (PINDER, inference-free). Per system we use the two AlphaFold-DB *predicted monomers* (pose-free; pLDDT in the B-factor column) as input and the *bound* complex only for labels. Each monomer is an intra-chain graph (residues within 8 Å of C_α); node features are per-residue **ESM2** embeddings (Lin et al., 2023), pLDDT, contact degree, and a chain indicator. The two monomers are joined by a *pose-free* cross-chain edge layer, the bottleneck knob: `none` (disconnected), `knn` (embedding-similarity bipartite edges), `fa` (fully-adjacent final layer), `virtual` (global hub), and `sdrf`, a *curvature-targeted* rewiring (Balanced-Forman, adding only the few edges that relieve the most negatively curved necks) (Topping et al., 2022). Interface labels (within 8 Å of the other chain in holo) are transferred onto predicted residues by *sequence alignment*. The predictor is a 4-layer GIN (64 hidden, LayerNorm + residuals), class-weighted CE, Adam (2×10^{-3}), 120 epochs; training is *deterministic* (TF32 off) and averaged over 10 seeds. We report AUPRC/AUROC/ECE and a base-rate-free *skill* = AUPRC − prevalence. We study heterodimers, skip degenerate systems (prevalence > 0.6 or < 0.02), and, being inference-free, test AlphaFold's interface *sub-task*, not end-to-end DockQ.

Two resistances and the richness sweep. The key design choice is *which* bottleneck to correlate with skill. Rather than the *native* resistance $R_{\text{eff}}^{\text{nat}}$ (holo contact graph, which the model never sees), we make the *modelled* resistance $R_{\text{eff}}^{\text{mod}}$ primary: the effective resistance *on the model's own pose-free input graph* between the true-interface residues, the bottleneck the network actually routes through ($R_{\text{eff}}^{\text{nat}}$ is kept as a contrast). We also run a high-power *within-system* test: per system we sweep the modelled width $k \in \{1,2,4,8\}$ and, centering both quantities within each system, correlate them, a paired design removing between-system noise. For the **headline experiment** we hold the GIN fixed and vary *only* the node features along a richness ladder: `baseline` (3-d), `onehot` (24-d), and ESM2 truncated to $\{32,128,480\}$-d; each regime *self-consistent* (its kNN layer rebuilt from its own features), reading off the rewiring benefit ΔAUPRC=AUPRC(`knn`)−AUPRC(`none`). Statistics use pre-registered *one-sided* Spearman tests, bootstrap CIs, and a minimum-detectable-effect 0.16 at n=243.

Hypotheses. **H0** (headline) the rewiring benefit depends on feature richness: $\Delta\text{AUPRC} > 0$ for weak features, < 0 for rich ones (over-squashing → over-smoothing). **H1** skill ↓ as $R_{\text{eff}}^{\text{mod}}$ ↑, tested cross- and within-system. **H2** cross-chain rewiring recovers accuracy (*causal*), with the trained-model Jacobian showing squashing is *relieved*. **H3** failures track *topology* more than AlphaFold's confidence (monomer pLDDT). **H4** flexibility (predicted→holo C_α-RMSD) and co-evolution (N_{eff}) are competing explanations. **H5** the link holds on the de-leaked *test* split.



(a) Rewiring benefit ΔAUPRC vs. feature richness: a sign-flip crossover.



(b) Cross-interface Jacobian on the trained models (weak regime).

Figure 1: Over-squashing on real data, and its limits. (a) The headline: the rewiring benefit flips from +0.086 to −0.017 AUPRC as features go weak→rich (over-squashing → over-smoothing). (b) In the weak regime the cross-interface gradient is 0 for the disconnected none (maximal squashing) and non-zero once cross edges carry interface signal, the mechanism behind Table 1.

3 Results

Synthetic probe: the mechanism is clean (H1/H2). As the bottleneck tightens ($k\downarrow$), A-node accuracy collapses toward chance, and the *same* two diagnostics we later carry to PINDER move in lockstep: across k, effective resistance anti-correlates with accuracy (ρ=−0.97) and the Jacobian correlates (ρ=+0.97). Depth does not rescue a tight bottleneck (the 2-layer curve never recovers; the 4-layer one needs $k \gtrsim 4$; Supplementary Fig. 3).

Headline (H0): over-squashing at the interface is contingent on feature richness. The textbook prediction (rewiring relieves the bottleneck, so it helps) turns out to hold only in a specific regime. Holding the GIN fixed and varying *only* the per-residue features along the richness ladder (each regime self-consistent), the rewiring benefit ΔAUPRC(knn−none) *crosses zero* (Fig. 1a): +0.086 (degree+pLDDT), +0.036 (one-hot), +0.001 (ESM-32), −0.008 (ESM-128), −0.017 (ESM-480). With weak features the model cannot identify interface residues locally, so it *must* route signal across the sparse interface, and the bottleneck over-squashes it; cross-chain edges relieve this and help. With rich ESM features each residue already carries enough evolutionary and structural context to be classified locally, so the same edges no longer relieve a needed bottleneck and instead mildly *over-smooth* the two chains together, hurting slightly. The interface bottleneck is thus real but *conditional*; it binds only when the task forces cross-chain routing. This both sharpens the thesis and locates the regime for the causal tests below, which we therefore run in the weak (onehot) feature regime.

Real H2: in the weak regime, cross-chain rewiring is causal and relieves squashing. Table 1 (the weak onehot regime) shows that *every* cross-chain rewiring beats the bottlenecked none baseline (AUROC 0.528→0.57–0.59; +0.03–0.04 AUPRC; ECE more than halved, 0.084→0.038), over 10 deterministic seeds. knn, fa, and sdrf add cross edges *without* a global hub, so their gains isolate *bottleneck relief*, not the virtual node's added capacity; the curvature-targeted sdrf matches them adding only ∼48 edges/system. The mechanism is confirmed directly on the trained models (Fig. 1b): the cross-interface Jacobian is identically 0 for the chain-disconnected

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Table 1: Real H2 (weak onehot regime). Interface predictor under five cross-chain edge layers (PINDER test, n=243; mean over 10 seeds; std $\leq$ 0.007 throughout). All cross-chain rewirings beat none; hub-free knn/fa/sdrf isolate bottleneck relief from the virtual node's added capacity.

rewiring	AUPRC	AUROC	ΔAUPRC	ECE
none	0.194	0.528	—	0.084
knn	0.232	0.589	+0.038	0.039
fa	0.224	0.566	+0.030	0.048
sdrf	0.230	0.584	+0.035	0.040
virtual	0.233	0.589	+0.039	0.038

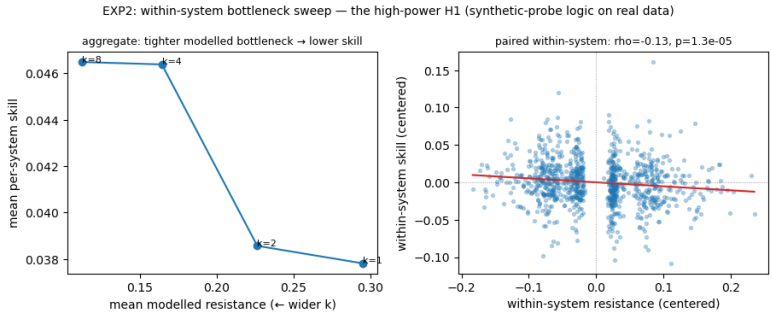


Figure 2: EXP2: within-system bottleneck sweep (the high-power H1). Left: aggregate per-system skill rises as the system's own modelled resistance falls (wider k). Right: centering both within each system gives a paired $\rho = -0.13$ (p=1.3×10^{-5}, 972 system–k points), the synthetic probe's logic on real data. The cross-system H1 scatter and the H3/H4 partial-correlation bars are in the (excluded) supplementary.

none (maximal squashing) and rises to 4.8×10^{-2} (knn), 5.8×10^{-2} (fa) once cross edges carry signal; rewiring relieves squashing, not merely adds parameters.

Real H1 / EXP2: the *modelled* bottleneck predicts failure (SUPPORTED). Aligning the diagnostic with the graph the model actually uses turns the first iteration's null into a strong signal. Cross-system, per-system skill anti-correlates with the *modelled* resistance at $\rho = -0.49$ (one-sided p=2×10^{-16}, 95% CI [−0.59, −0.39], sign-stable across all 10 seeds), whereas the *native* (pose-unseen) resistance stays weak ($\rho = -0.11$, p=0.075), as it must, since the model never sees the holo geometry. A *within-system* sweep confirms it at high power: sweeping the modelled width k and centering within each system gives a paired $\rho = -0.13$ (p=1.3×10^{-5}, CI [−0.19, −0.07], 972 system–k points), with skill falling as the system's own bottleneck tightens in 57% of systems (Fig. 2) , the synthetic probe's logic reproduced on real data.

Real H3/H4: topology beats confidence (SUPPORTED). In partial Spearman correlations (each term controlling for the others), the modelled-resistance term ($\rho = -0.47$, p=5.6×10^{-15}) is an order of magnitude larger than the strongest confidence term, monomer pLDDT ($|\rho|$=0.075), and dwarfs flexibility and co-evolution N_{eff} (both $|\rho|$<0.06, p>0.3; Supplementary Fig. 4), so H4's competitors are informative in the negative. One caveat: PINDER did not expose AlphaFold-Multimer's *complex* confidence (ipTM/PAE), so our baseline is the weaker *monomer* pLDDT; at n=243 the minimum detectable $|\rho|$ is 0.16, so small terms read as bounded, not absent.

Real H5: generalisation gradient (DIRECTIONAL). Stratifying *test* skill by *modelled*-resistance tertile (low→high) gives a monotone decline ($0.073 \to 0.035 \to 0.025$), but with only ∼81 systems/tertile and no trend test we read this as *directional*, not confirmed; the powered continuous within-split H1 is the same claim done properly and *is* significant, tying over-squashing to OOD generalisation. The large train→test skill drop reflects expected overfitting of a small GIN on 383 systems.

4 Relation to the literature and course concepts

This study operationalises several course concepts. *Over-squashing* and *information propagation* (Alon & Yahav, 2021) are measured directly via *effective resistance* (Black et al., 2023) and Jacobian sensitivity (Topping et al., 2022; Di Giovanni et al., 2023); *graph rewiring* as an inductive-bias intervention is the remedy literature (virtual nodes, fully-adjacent layers, curvature/spectral rewiring such as SDRF (Topping et al., 2022), BORF (Nguyen et al., 2023), FOSR (Karhadkar et al., 2023), plus expander propagation, EGP (Deac et al., 2022)), of which our H2/SDRF is a domain instance, while the *expressive-power* view (Di Giovanni et al., 2023) explains why depth alone cannot rescue a tight neck. Our crossover places *over-squashing* and *over-smoothing* (NT & Maehara, 2019; Rusch et al., 2023) on one feature-richness axis. Finally, PINDER's *leakage-free* split (Kovtun et al., 2024) frames the question as *generalisation under shift*, and we reframe the AlphaFold-Multimer "confident monomers, wrong complex" failure (Evans et al., 2022; Bryant et al., 2022) (cf. interface predictors like MaSIF (Gainza et al., 2020)) as a GNN information-flow problem.

5 Discussion, critique, and outlook

The central lesson is that over-squashing at the interface is *real but contingent*, and the crossover maps when it binds: weak features force cross-chain routing and *over-squashing* dominates (rewiring helps); rich ESM features make the interface locally solvable and the same edges *over-smooth* (rewiring hurts). The textbook fix is therefore regime-dependent, more honest than "rewiring always helps." In the weak regime the evidence is coherent (causal: hub-free rewirings and a parsimonious SDRF rule out pure capacity, and the Jacobian shows squashing relieved; observational: the *modelled* bottleneck tracks failure at $\rho = -0.49$); the one dissociation (the pose-unseen *native* resistance is weak) is principled for *pose-free* inputs. A scope note sharpens the contribution: we show over-squashing in a message-passing predictor we control, *not* inside AlphaFold-Multimer, whose all-to-all attention is the bottleneck-*relieving* (fully-adjacent) regime our crossover identifies; so our own mechanism predicts dense attention should *escape* this failure, framing over-squashing as the price of the sparse, scalable models used at interface scale.

We are candid about limits: the absolute predictor is weak (test AUROC 0.589, skill $\approx +0.04$), so claims rest on *relative* effects exceeding their 10-seed std, not a strong classifier; weak correlations are reported with bootstrap CIs, one-sided tests, and an MDE so they read as bounded, not headline. A caveat attaches to H1: *modelled* resistance is read off the model's own input graph, so its cross-system tie to skill partly shares a dependence on feature informativeness; the within-system sweep (EXP2) is the cleaner, confound-resistant test we lean on. Future work should use *pose-aware*, *edge-level* inputs and, given per-system AF2-Multimer DockQ, test the thesis on AlphaFold's true output, not its sub-task.

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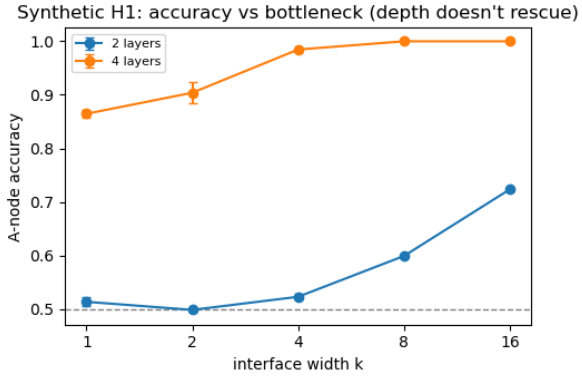
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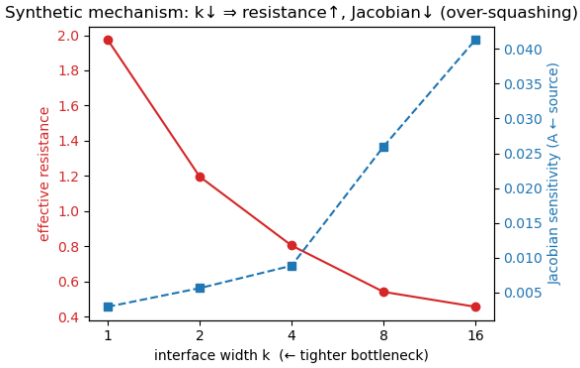
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Supplementary material

A. Synthetic probe. Two Erdős–Rényi communities (A, B; 16 nodes each) joined by exactly k cross-edges; a single B-source label is read out at the A nodes, so the task requires cross-interface flow. Figure 3 reports A-node accuracy, cross-interface R_{eff}, and Jacobian sensitivity across $k \in \{1, 2, 4, 8, 16\}$ and depth $\in \{2, 4\}$.

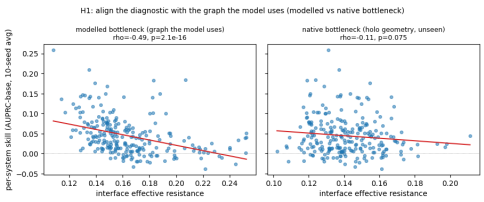


(a) Accuracy vs. bottleneck width k.

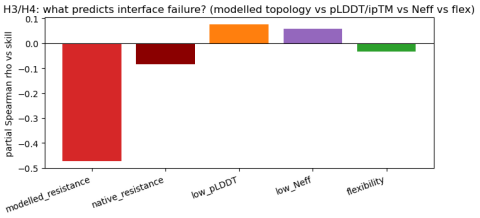


(b) Resistance ↑, Jacobian ↓ as k↓.

Figure 3: Synthetic probe. A controllable interface bottleneck reproduces over-squashing: accuracy collapses as k tightens, quantitatively tracked by effective resistance and Jacobian sensitivity (ρ=−0.97/+0.97 vs. accuracy), the metrics we carry to PINDER.



(a) H1: modelled (left) vs. native (right) resistance vs. skill, cross-system.



(b) H3/H4: partial-correlation predictors of failure.

Figure 4: Cross-system predictors. (a) The *modelled* bottleneck out-tracks the pose-unseen *native* one. (b) Modelled topology dwarfs confidence (pLDDT), flexibility, and N_{eff}.

B. Cross-system H1 and H3/H4 predictors.

C. Hypothesis scorecard. Table 2 collects the predicted-vs-observed verdict for every hypothesis (each also stated in the running text of §3).

D. Implementation. GIN: 4 layers, 64 hidden, LayerNorm + residuals; class-weighted cross-entropy (positive weight capped at 3); Adam (2×10^{-3}, weight decay 10^{-5}), 120 epochs; deterministic training (TF32 off, deterministic algorithms), 10 seeds. ESM2 features from `esm2_t12_35M_UR50D` (480-d), cached per system. Full code, the executed notebook, and the result tables/CSVs are in the repository linked above.

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Table 2: Hypothesis scorecard. Predicted vs. observed (PINDER test, n=243, 10 seeds).

	prediction	observed	verdict
H0	rewiring benefit depends on features	ΔAUPRC flips $+0.086 \rightarrow -0.017$	**supported (crossover)**
H1	skill $\downarrow$ as $R_{\text{eff}}^{\text{mod}} \uparrow$	$\rho = -0.49$ cross-sys; -0.13 within	**supported**
H2	rewiring recovers accuracy	all knobs >`none`; Jacobian relieved	**supported (causal)**
H3	topology > confidence	modelled -0.47 vs. pLDDT 0.075	**supported**
H4	flexibility/N_{eff} compete	neither predicts failure	negative (informative)
H5	link holds on test split	monotone test tertiles ($n\sim81$)	directional

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